Meridian Genetics Laboratory

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MOLECULAR GENETICS — DIAGNOSTIC EXOME, TARGETED ANALYSIS

Patient:	Doe, Jordan (fictional)	**Accession:**	MG-2026-004821
DOB / Sex:	2021-08-14 / M	**Specimen:**	Peripheral blood (EDTA)
MRN:	FIC-0099231	**Collected:**	2026-05-02
Ordering provider:	S. Nakamura, MD (Child Neurology)	**Reported:**	2026-05-29

CLINICAL HISTORY

4-year-old boy referred for genetic evaluation of a complex neurodevelopmental and connective-tissue presentation. History of recurrent febrile and afebrile seizures with onset at 6 months of age, now with global developmental delay and gait ataxia. On ophthalmologic exam, bilateral ectopia lentis (lens dislocation) was noted. Echocardiogram demonstrated dilatation of the aortic root. Tall stature with arachnodactyly. No reported family history of sudden cardiac death. Testing requested to identify a molecular cause.

RESULTS — REPORTED VARIANTS

Gene	Variant (cDNA)	Protein	Zygosity	Inheritance	Classification
SCN1A	c.3637C>T	p.(Arg1213Ter)	Heterozygous	de novo	Pathogenic
FBN1	c.4082G>A	p.(Cys1361Tyr)	Heterozygous	AD	Likely pathogenic
MYH7	c.1063G>A	p.(Ala355Thr)	Heterozygous	AD	Uncertain significance

INTERPRETATION

SCN1A c.3637C>T (p.Arg1213Ter) is a nonsense variant classified as **Pathogenic**. Loss-of-function variants in SCN1A cause Dravet syndrome (developmental and epileptic encephalopathy), consistent with this patient's early-onset seizures, developmental delay, and ataxia.

FBN1 c.4082G>A (p.Cys1361Tyr) is a missense variant affecting a conserved cysteine and is classified as **Likely pathogenic**. FBN1 variants cause Marfan syndrome; the patient's ectopia lentis and aortic root dilatation are characteristic and are not explained by the SCN1A finding.

MYH7 c.1063G>A (p.Ala355Thr) is classified as a **variant of uncertain significance**. Its clinical relevance is unknown and it should not be used for medical management at this time.